

Quiz 6

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Let $A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

(a) Reduce $[A \ b]$ to $[R \ d]$ (the reduced echelon form) to find the condition on b_1, b_2, b_3 such that $Ax = b$ is solvable.

$$[A \ b] = \begin{bmatrix} 1 & 3 & 1 & 2 & | & b_1 \\ 2 & 6 & 4 & 8 & | & b_2 \\ -2 & -6 & 0 & 0 & | & b_3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 & | & b_1 \\ 0 & 0 & 2 & 4 & | & b_2 - 2b_1 \\ 0 & 0 & 2 & 4 & | & b_3 + 2b_1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 & | & b_1 \\ 0 & 0 & 2 & 4 & | & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & | & b_3 - b_2 + 4b_1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 3 & 0 & 0 & | & 2b_1 - \frac{1}{2}b_2 \\ 0 & 0 & 2 & 4 & | & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & | & b_3 - b_2 + 4b_1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 3 & 0 & 0 & | & 2b_1 - \frac{1}{2}b_2 \\ 0 & 0 & 1 & 2 & | & \frac{1}{2}b_2 - b_1 \\ 0 & 0 & 0 & 0 & | & b_3 - b_2 + 4b_1 \end{bmatrix} = [R \ d]$$

$$\begin{matrix} \downarrow & \downarrow \\ R & d \end{matrix}$$

$$\Rightarrow Ax = b \text{ is solvable iff } \underline{b_3 - b_2 + 4b_1 = 0}$$

(b) Find the complete solution to $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$.

$$\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$$

$$[\mathbf{A} \ \mathbf{b}] \rightarrow [\mathbf{R} \ \mathbf{d}] = \begin{bmatrix} 1 & 3 & 0 & 0 & | & \frac{1}{2} \\ 0 & 0 & 1 & 2 & | & \frac{1}{2} \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix} \Rightarrow \text{Free variables: } x_2, x_4 \\ \text{pivot variables: } x_1, x_3$$

Find $x_p \rightarrow$ solve $\mathbf{R}x_p = \mathbf{d}$.

$$\text{set } x_2 = x_4 = 0 \Rightarrow \begin{cases} x_1 = \frac{1}{2} \\ x_3 = \frac{1}{2} \end{cases} \Rightarrow x_p = \begin{bmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \\ 0 \end{bmatrix}$$

Find $x_n \rightarrow$ solve $\mathbf{R}x_n = 0$

$$\text{set } x_2 = 1, x_4 = 0 \Rightarrow \begin{cases} x_1 + 3 = 0 \\ x_3 = 0 \end{cases} \Rightarrow x_1 = -3, x_3 = 0 \Rightarrow s_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{set } x_2 = 0, x_4 = 1 \Rightarrow \begin{cases} x_1 = 0 \\ x_3 + 2 = 0 \end{cases} \Rightarrow x_1 = 0, x_3 = -2 \Rightarrow s_2 = \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

Thus

$$\mathbf{x} = x_p + C_1 s_1 + C_2 s_2$$

$$= \begin{bmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} + C_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + C_2 \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$